

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-16 21:44 UTC

## SEC P vs NP — notebook summary

Generated 2026-05-16 21:44 UTC

- Cycles recorded: **714**
- Time span: 552.1h (~1.29 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 677   |
| BARRIER_HIT  | 18    |
| FALSIFIED    | 15    |
| SUPPORTED    | 4     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Real Algebraic Geometry                                                           | 40     |
| Schur-Weyl Duality                                                                | 28     |
| Free Probability Theory                                                           | 19     |
| Representation Theory of Symmetric Groups                                         | 18     |
| Matroid Theory                                                                    | 16     |
| Free Probability                                                                  | 14     |
| Additive Combinatorics                                                            | 13     |
| Noncommutative $L^p$ Geometry                                                     | 10     |
| Random Matrix Theory                                                              | 8      |
| Polymatroid Theory                                                                | 8      |
| Algebraic Geometry                                                                | 7      |
| Spectral Graph Theory                                                             | 7      |
| Finite Geometry                                                                   | 6      |
| Plethysm Theory                                                                   | 6      |
| Persistent Homology                                                               | 5      |
| Noncommutative Harmonic Analysis                                                  | 5      |
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier Analysis on Boolean Functions                                             | 4      |

| Field                                                                                              | Cycles |
|----------------------------------------------------------------------------------------------------|--------|
| Algebraic Geometry over Finite Fields                                                              | 4      |
| Fourier Analysis of Boolean Functions                                                              | 4      |
| Algebraic Geometry of Secant Varieties                                                             | 4      |
| FOURIER_ANALYSIS                                                                                   | 4      |
| Noncommutative Geometry                                                                            | 4      |
| Fourier analysis of boolean functions                                                              | 3      |
| Algebraic Topology                                                                                 | 3      |
| matroid theory                                                                                     | 3      |
| Coarse Geometric Lifting (CGL) — Geometric Complexity Theory                                       | 3      |
| {“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”} | 3      |
| Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)                                              | 3      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                             |
|----------------------|--------------|-------------------------------------------------------------------|
| 2026-05-13 06:36 UTC | INCONCLUSIVE | Hilbert Polynomial Leading Coefficient Inversely Proportional to  |
| 2026-05-13 06:42 UTC | INCONCLUSIVE | Kronecker Coefficient Gap in Symmetric Powers of                  |
| 2026-05-13 07:16 UTC | INCONCLUSIVE | Permanent vs Det                                                  |
| 2026-05-13 07:23 UTC | INCONCLUSIVE | SOS Moment Matrix Eigenvalue Sum Bounded by Degree for Max-CUT    |
| 2026-05-13 08:49 UTC | INCONCLUSIVE | Symmetric Group Fourier Coefficient Count Inversely Proportional  |
| 2026-05-13 09:19 UTC | INCONCLUSIVE | Free Cumulant Sum Lower Bounded by Communication Complexity of DI |
| 2026-05-13 09:31 UTC | INCONCLUSIVE | Schur-Weyl Component Count Gap in Monotone Circuit Size for Perma |
| 2026-05-13 09:59 UTC | INCONCLUSIVE | Algebraic Shifting Generator Count Bounds Communication Complexit |
| 2026-05-13 11:20 UTC | INCONCLUSIVE | Algebraic Shifting Edge Count Bounds Monotone DNF Size for k-CLIQ |
| 2026-05-13 11:47 UTC | INCONCLUSIVE | Symmetric Square Irreducible Component Count Gap in Permanent vs  |
| 2026-05-13 12:08 UTC | INCONCLUSIVE | Symplectic Capacity Inverse Proportional to Extended Frege Proof  |
|                      |              | Symmetric Decomposition Coefficient Count Inversely Proportional  |

| Time                 | Verdict      | Title                                                             |
|----------------------|--------------|-------------------------------------------------------------------|
| 2026-05-16 20:13 UTC | INCONCLUSIVE | Layer-Commutator Frobenius Discrepancy Separates Read-Twice BP fo |
| 2026-05-16 20:46 UTC | INCONCLUSIVE | Matroid Matching of Term Co-Occurrence Graph Bounds k-CLIQUE Mono |
| 2026-05-16 21:14 UTC | INCONCLUSIVE | Persistent H1 of Random Row Subclouds Bounds DISJ Communication   |
| 2026-05-16 21:44 UTC | INCONCLUSIVE | Non-Backtracking Spectral Gap Bounds Tseitin Resolution Length    |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```